Road Damage and Accident Risk Analytics Dashboard using Business Intelligence

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Abstract:

Road infrastructure monitoring and maintenance play a critical role in ensuring transportation safety and urban development. Inefficient repair tracking, unmanaged road damage, and delayed maintenance responses contribute significantly to accident risk and economic loss. This study presents a comprehensive visual analytics framework developed using Power BI to analyze road damage, accident patterns, repair status, and administrative-level infrastructure performance across municipalities and panchayat regions. The proposed interactive dashboard integrates multi-dimensional data attributes including traffic density, weather conditions, vehicle speed limits, accident history, repair status, and local governance units. Key analytical components include accident risk correlation, weather–traffic interaction matrices, vehicle speed risk indicators, geographical mapping of high-risk zones, and repair status monitoring across municipalities and rural panchayat regions. The system enables comparative analysis between repaired and pending road segments, identifies accident-prone administrative regions, and highlights maintenance gaps in urban and rural areas. Interactive slicers and conditional formatting enhance decision-making by allowing dynamic filtering across traffic level, weather conditions, and repair categories. The analysis reveals that accident frequency is significantly influenced by the combined impact of high traffic density, adverse weather conditions, and elevated speed limits. Furthermore, regions with delayed repair status exhibit higher risk concentration, emphasizing the importance of proactive infrastructure management. This study demonstrates how business intelligence tools can transform structured transportation datasets into actionable decision-support systems. The developed framework supports data-driven planning for municipalities and panchayat authorities, contributing to improved road safety, efficient maintenance allocation, and sustainable infrastructure governance.

Keywords:

Road Infrastructure Analytics – Power BI Dashboard – Business Intelligence System – Accident Risk Assessment – Road Damage Monitoring – Repair Status Tracking – Municipal Infrastructure Management – Panchayat-Level Governance Analytics – Traffic Density Analysis – Weather Impact Assessment – Vehicle Speed Risk Evaluation – Geospatial Risk Mapping – Interactive Data Visualization – Conditional Formatting Analysis – KPI-Based Monitoring – Data-Driven Decision Support – Transportation Safety Management – Urban and Rural Infrastructure Planning – Digital Infrastructure Governance – Smart City Analytics.

I. INTRODUCTION

Road infrastructure is a fundamental component of socio-economic development, enabling mobility, trade, emergency response, and regional connectivity. The efficiency and safety of transportation systems depend heavily on the structural integrity and maintenance quality of road networks. However, rapid urban expansion, increasing vehicular density, environmental stressors, and administrative resource constraints have intensified the challenges associated with road monitoring and maintenance management. Inadequate infrastructure surveillance, delayed repair interventions, and fragmented reporting systems contribute significantly to accident risks, operational inefficiencies, and economic losses across both urban municipalities and rural panchayat regions.

Road damage, including surface cracks, potholes, and structural deterioration, directly impacts traffic flow and vehicle stability. Studies indicate that accident frequency often correlates with poor road conditions, adverse weather patterns, and excessive speed levels. Despite the availability of structured transportation

datasets, many local governance bodies lack comprehensive analytical systems capable of transforming raw data into actionable insights. Traditional monitoring approaches rely on manual inspections and static reports, which are time-consuming, error-prone, and insufficient for dynamic decision-making.

Recent advancements in Business Intelligence (BI) technologies have enabled the development of interactive, scalable, and data-driven monitoring frameworks. BI platforms facilitate multi-source data integration, advanced modeling, real-time filtering, and interactive visualization, thereby enhancing transparency and accountability in public infrastructure management. By incorporating key performance indicators (KPIs), correlation analysis, and geospatial mapping techniques, decision-makers can identify critical risk zones, evaluate infrastructure performance, and prioritize maintenance allocation effectively.

This research presents a comprehensive Road Infrastructure Analytics Framework developed using Power BI to support data-driven governance and transportation safety management. The proposed system integrates multiple data dimensions, including traffic density, weather conditions, vehicle speed limits, accident history, repair status, and administrative segmentation at municipal and panchayat levels. The analytical model incorporates accident risk correlation matrices, weather–traffic interaction analysis, vehicle speed risk indicators, geospatial hotspot mapping, and repair performance tracking dashboards.

The framework enables comparative assessment between repaired and pending road segments, highlights accident-prone administrative clusters, and identifies maintenance inefficiencies across urban and rural jurisdictions. Interactive slicers, drill-through features, conditional formatting mechanisms, and KPI-based monitoring enhance analytical flexibility and improve stakeholder engagement. Furthermore, the system supports scenario-based evaluation by dynamically filtering traffic intensity, weather categories, and repair classifications.

The findings derived from the analytical model indicate that accident frequency is significantly influenced by the combined interaction of high traffic density, unfavorable weather conditions, and elevated speed limits. Regions with delayed repair status demonstrate concentrated risk patterns, emphasizing the necessity of proactive and predictive infrastructure management strategies.

By transforming structured transportation datasets into an integrated visual decision-support environment, this study demonstrates the practical application of business intelligence tools in enhancing road safety, optimizing maintenance resource allocation, and strengthening digital governance frameworks. The proposed approach contributes to sustainable infrastructure planning and supports the evolution of smart city and rural digital transformation initiatives.

II. RELATED WORK

Road damage and pothole detection have been extensively studied using computer vision and deep learning techniques. Several researchers have applied Convolutional Neural Networks (CNNs), transfer learning models, and image classification frameworks to automatically identify cracks and potholes from road surface images. These AI-based systems significantly reduce manual inspection effort and improve detection accuracy.

In addition to vision-based detection, recent studies have emphasized the importance of transportation analytics for accident risk assessment. Multi-factor analysis incorporating traffic density, weather conditions, vehicle speed limits, and repair status has been used to understand accident patterns and infrastructure performance. However, many of these works focus either on image-based detection models or predictive machine learning approaches without integrating administrative level governance analytics.

Business Intelligence (BI) platforms such as Power BI have emerged as powerful tools for structured data visualization, KPI monitoring, and interactive decision-support systems. Despite their capability to transform transportation datasets into actionable insights, limited research integrates road damage monitoring, accident correlation analysis, geospatial hotspot identification, and repair tracking into a unified dashboard framework.

This study addresses this gap by developing a comprehensive Power BI-based analytical dashboard to support AI-driven road damage monitoring and infrastructure risk analysis. The proposed framework integrates multi-dimensional transportation attributes into a unified Business Intelligence environment for systematic accident and maintenance evaluation. It enables administrative-level comparison across municipalities and panchayat regions to support transparent governance. Geospatial visualization and

KPI-based monitoring are incorporated to identify high-risk clusters and delayed repair segments. The framework enhances data-driven planning and strategic road safety management.

III. PROPOSED METHODOLOGY

The proposed methodology focuses on transforming structured road infrastructure data into an interactive visual decision-support framework using Power BI.

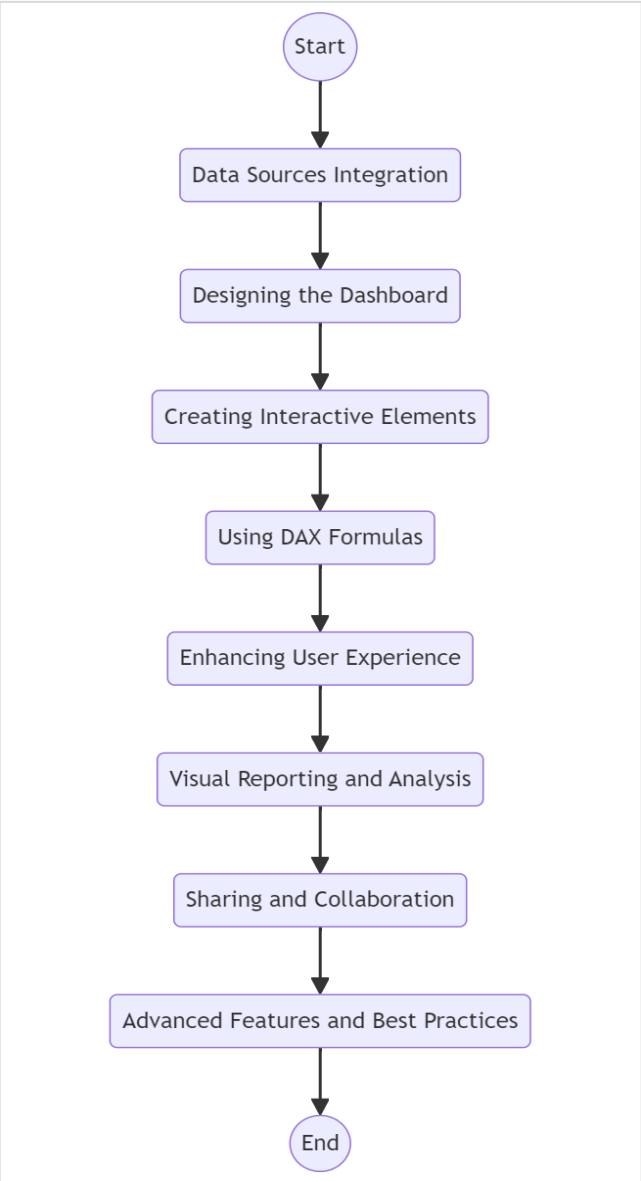

Fig 1 Power BI Implementation Architecture

A. Data Acquisition and Integration

The first stage involves the collection and integration of structured transportation and infrastructure datasets. The dataset includes attributes such as city and state identifiers, traffic density levels, weather conditions, vehicle speed limits, accident counts, road damage counts, repair status, and administrative classification (municipality or panchayat).

Data from multiple sources were imported into the Power BI environment and consolidated into a unified schema to ensure consistency and compatibility for analytical processing.

B. Data Preprocessing and Transformation

Data preprocessing was performed using the Power Query Editor. This step ensures data quality and analytical readiness. The following operations were carried out:

- Removal of duplicate and
- Inconsistent records
- Handling of missing values
- Standardization of categorical variables
- Data type corrections
- Creation of derived analytical attributes

Feature engineering techniques were applied to generate new variables such as accident intensity levels and repair delay indicators. These transformations improve analytical depth and enable multidimensional risk evaluation.

C. Data Modeling and KPI Definition

A structured relational data model was established to support efficient filtering and cross-dimensional analysis. Appropriate relationships were defined between fact and dimension tables to enable hierarchical drill-down analysis.

Key Performance Indicators (KPIs) were defined using Data Analysis Expressions (DAX). The primary KPIs include:

- Total Accident Count
- Total Road Damage Instances
- Repair Completion Ratio
- Administrative Risk Distribution
- Accident Risk Index

These measures serve as quantitative indicators for evaluating infrastructure performance and safety conditions across administrative regions.

D. DAX-Based Analytical Computation

Advanced analytical measures were implemented using DAX functions to compute aggregated metrics, conditional risk indicators, and comparative performance measures. DAX formulas were used to:

- Calculate dynamic accident rates
- Compute repair status percentages
- Derive risk indices based on traffic and weather conditions
- Enable time-based trend analysis

This stage enhances the computational capability of the system and supports real-time interactive filtering.

E. Interactive Dashboard Development

An interactive dashboard was designed to visually represent analytical findings. Multiple visualization components were integrated, including:

- KPI summary cards
- Geospatial risk mapping
- Traffic–weather interaction matrices
- Time-series trend charts
- Administrative-level comparison charts

Interactive slicers and drill-through capabilities were implemented to allow users to dynamically explore specific regions, traffic conditions, and repair categories.

F. Risk Analysis and Visualization

Multidimensional risk analysis was performed to examine the relationship between accident frequency and contributing factors such as traffic density, weather conditions, speed limits, and repair status. Visual analytical tools were used to detect high-risk clusters and identify accident-prone administrative zones.

The system enables comparative evaluation between repaired and pending road segments, highlighting maintenance gaps and infrastructure inefficiencies.

G. Deployment and Stakeholder Integration

The finalized dashboard was deployed through Power BI Service to enable accessibility for decision-makers and administrative authorities. The system supports collaborative sharing and periodic performance monitoring.

By integrating structured data processing with interactive visualization, the proposed methodology establishes a scalable Business Intelligence framework for data-driven infrastructure governance and transportation safety management.

IV. EXPERIMENTAL SETUP

The experimental setup was designed to evaluate the performance of the Power BI-based analytical framework using a structured transportation dataset containing road infrastructure records across multiple regions. The setup includes dataset configuration, implementation environment, data modeling architecture, and analytical evaluation criteria.

A. Dataset Description

The experimental analysis was conducted using a structured dataset containing road infrastructure records across multiple cities and states in India. The dataset includes attributes such as traffic density, weather conditions, vehicle speed limits, accident count, road damage instances, repair status, latitude and longitude coordinates, and administrative classification (municipality and panchayat regions).

These multi-dimensional attributes enable comprehensive accident risk analysis, hotspot identification, and infrastructure performance evaluation at both urban and rural administrative levels.

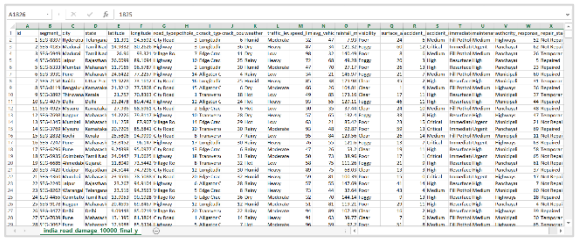

Fig 2 Sample Dataset Records

B. Implementation Environment

The analytical framework was implemented using Microsoft Power BI Desktop as the primary Business Intelligence platform. Data preprocessing and transformation were performed using Power Query Editor to ensure consistency and analytical readiness.

DAX (Data Analysis Expressions) functions were used to compute key performance measures, and the finalized dashboard was deployed through Power BI Service for interactive stakeholder access.

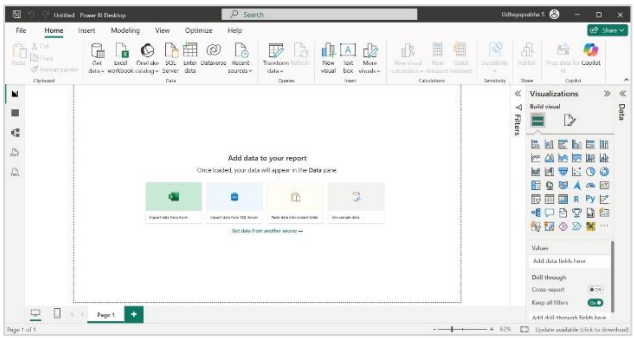

Fig 3 Power BI Development Environment

C. Data Modeling Architecture

A structured relational model was developed to enable efficient cross-filtering and hierarchical analysis. Fact tables containing accident and road damage metrics were connected to dimension tables representing location, traffic, weather, and administrative categories.

Calculated measures were created to derive KPIs such as total accidents, total damages, accident risk index, repair completion ratio, and city-level accident intensity.

D. Visualization and Analytical Components

The experimental dashboard consists of multiple analytical pages including overview KPIs, city-level analysis, time trend analysis, geographical accident hotspot mapping, and repair status monitoring.

Visualization techniques include KPI summary cards, geospatial heat maps using latitude–longitude data, box-and-whisker plots for accident distribution analysis, correlation matrices, and conditional formatting for risk highlighting.

E. Evaluation Criteria

The evaluation focused on identifying high-risk zones, analyzing traffic–weather interactions, and examining the impact of delayed repairs on accident concentration.

Comparative analysis was performed between municipalities and panchayat regions to detect infrastructure governance disparities. The experimental results validate the capability of the proposed dashboard to support data-driven road safety monitoring and infrastructure decision-making.

V. RESULTS AND DISCUSSION

The proposed Power BI-based analytical framework was evaluated using structured transportation and road infrastructure data. The dashboard modules provide multi-dimensional insights into road damage distribution, accident risk patterns, administrative-level performance, and infrastructure maintenance gaps. The results obtained from the implemented analytical pages are discussed below.

A. Executive Overview Analysis

As shown in Fig 4, the Executive Overview dashboard presents consolidated KPIs including total roads, total accidents, total cracks, total potholes, and response time indicators. The aggregated metrics reveal significant variation in infrastructure conditions across regions.

The accident risk category visualization indicates that a considerable proportion of road segments fall under high and critical risk classifications, highlighting the need for prioritized intervention. This overview enables rapid assessment of infrastructure health and supports strategic decision-making at administrative levels.

Fig 4 Overview KPIs

B. Road Damage Distribution Analysis

The Road Damage Analysis module (Fig 5) provides state-wise and city-wise distribution of potholes and cracks. The results show that damage intensity varies significantly across cities, indicating uneven maintenance effectiveness and traffic load distribution.

Crack type percentage analysis reveals a nearly balanced distribution among crack categories, suggesting that deterioration patterns are influenced by multiple structural and environmental factors. Authority-wise management comparison further

indicates disparities in infrastructure handling between panchayat, municipal, and highway departments.

Fig 5 Damage Distribution

C. Accident Risk Analysis

The Accident Risk Analysis dashboard (Fig 6) evaluates the relationship between accident frequency and contributing variables such as traffic density, weather conditions, and vehicle speed limits. The traffic density correlation analysis indicates higher accident concentration under heavy traffic conditions.

Weather-based filtering demonstrates noticeable variation in accident counts across climatic conditions, confirming the multi-factor nature of accident occurrence. The geographical hotspot map visually identifies accident-prone clusters, enabling location-specific intervention planning.

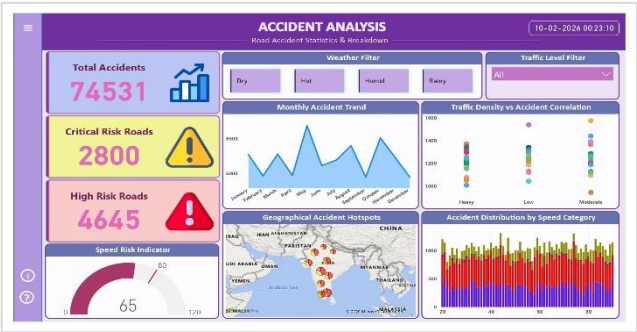

Fig 6 Risk Analysis

D. Navigation and Maintenance Analysis

The developed dashboard incorporates a structured multi-page navigation mechanism to enable seamless analytical exploration across functional modules. As illustrated Fig 7, the navigation interface supports dynamic switching between executive overview, damage analysis, accident risk assessment, and maintenance monitoring modules. Interactive slicers and drill-through capabilities facilitate filtered evaluation based on city, traffic density, weather conditions, administrative authority, and repair status.

The Maintenance Analysis module (Fig 8) provides comparative insights into repaired and pending road segments across administrative regions. The analytical framework identifies maintenance backlogs and highlights risk concentration in delayed repair zones, thereby supporting proactive infrastructure planning and governance-level decision-making.

In addition to the presented modules, the framework supports city-level comparative analysis and temporal trend evaluation to examine accident growth patterns and infrastructure performance over time. These analytical capabilities strengthen the decision-support effectiveness of the proposed system by enabling region-specific prioritization and time-based infrastructure monitoring.

The integrated architecture enhances usability, promotes scenario-driven analysis, and ensures that stakeholders can derive actionable insights without requiring advanced technical expertise.

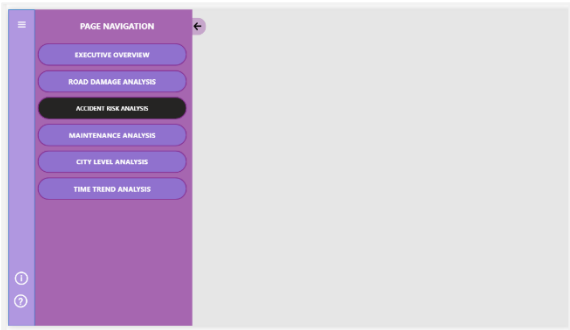

Fig 7 Page Navigation

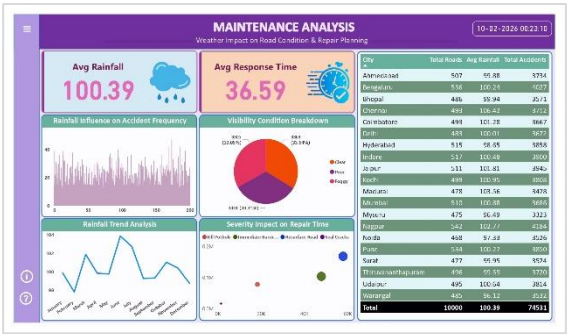

Fig 8 Maintenance Analysis Interface

E. Discussion

The experimental results validate the effectiveness of the proposed Business Intelligence framework in transforming structured transportation data into actionable insights. The integration of KPI

monitoring, geospatial hotspot identification, and multi-factor correlation analysis enables comprehensive infrastructure evaluation.

The findings indicate that accident risk is strongly influenced by traffic density, environmental conditions, and delayed repair status. Additionally, administrative-level disparities suggest the need for improved governance coordination and proactive maintenance strategies.

Overall, the proposed dashboard demonstrates the potential of Business Intelligence tools in supporting data-driven road safety management and infrastructure planning within smart city ecosystems.

VI. CONCLUSION

This study presented a structured analytical framework for AI-powered road damage and pothole monitoring through a Power BI-based visual intelligence system. The proposed dashboard integrates multi-dimensional transportation attributes, including accident frequency, traffic density, weather conditions, repair status, administrative authority, and geographic location, into a unified decision-support platform. By transforming structured datasets into interactive visual representations, the system enables comprehensive infrastructure performance evaluation.

The analytical results demonstrate that accident occurrence is not influenced by isolated factors but rather by the combined impact of high traffic density, adverse environmental conditions, and delayed maintenance responses. City-level comparative evaluation revealed noticeable disparities in infrastructure stress and administrative performance, while geospatial hotspot identification highlighted accident-prone regions requiring immediate intervention. Maintenance analysis further emphasized the correlation between pending repair status and elevated risk concentration.

The developed framework enhances governance-level transparency by enabling KPI-based monitoring, regional prioritization, and scenario-driven analysis. Unlike conventional static reporting systems, the proposed interactive dashboard supports dynamic filtering and drill-through exploration, allowing stakeholders to derive targeted insights efficiently.

Overall, the study validates the effectiveness of Business Intelligence tools in supporting data-driven transportation safety management and infrastructure planning. The proposed system contributes toward proactive maintenance allocation, optimized resource distribution, and sustainable urban and rural infrastructure governance, thereby strengthening smart city decision-support ecosystems.

VII. REFERENCES

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